Homework — 1.4.3 Boolean Algebra — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Total: Part A 14 + Part B 6 = 20 marks.

Part A

1. [3] Q = (A + B) · ¬C:

A	B	C	A+B	¬C	Q
0	0	0	0	1	**0**
0	0	1	0	0	**0**
0	1	0	1	1	**1**
0	1	1	1	0	**0**
1	0	0	1	1	**1**
1	0	1	1	0	**0**
1	1	0	1	1	**1**
1	1	1	1	0	**0**

Mark: A+B column correct (1); ¬C column correct (1); Q column correct — Q = 1 only when (A OR B) is true AND C = 0 (1).

2. [2] Simplify `¬(A · ¬B)`: - **De Morgan's law** applied to the outer bar: `¬(A · ¬B) = ¬A + ¬(¬B)` (break the bar, change · to +, negate each term) (1). - **Double negation**: `¬(¬B) = B`, giving the result **`¬A + B`** (1).

3. [2] Simplify `A + ¬A · B`: - Result: **`A + B`** (1). - Valid law-based working/naming (1), e.g. **distribution**: `A + ¬A·B = (A + ¬A) · (A + B)`; `A + ¬A = 1` (complement); `1 · (A + B) = A + B` (identity). *(Quoting the redundancy identity `A + ¬A·B = A + B` with the laws named also gains this mark.)*

4. (a) [1] Completed Karnaugh map (all eight cells correct):

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	1	1	0	0
A = 1	1	1	1	0

(b) [3] - Group of **four** covering both rows of the BC = 00 and BC = 01 columns → **¬B** (1). - Group of **two** in the A = 1 row covering BC = 01 and BC = 11 → **A·C** (1). - Simplified expression: **X = ¬B + A·C** (1).

Groups must be rectangles whose sizes are powers of two; groups may overlap (the cell A=1, BC=01 belongs to both). Verified against the truth table: ¬B + A·C = 1 exactly on rows 000, 001, 100, 101, 111.

5. [2] Half adder: **Sum = A ⊕ B**, **Carry = A · B** (1 — both correct). It cannot be used alone for the middle columns because a half adder has **only two inputs (A and B) and no carry-in**, so it cannot accept the **carry produced by the previous column** — the middle and higher columns of a multi-bit addition each need to add three bits (A, B and the carry-in), which requires a **full adder** (1). *(AO1)*

6. [1] A D-type flip-flop stores a **single bit** (0 or 1), and its output can only change on a **clock edge/pulse** — the stored value is held constant between clock edges (both ideas needed for the mark). *(AO1)*

Part B

7. [3] Symmetric encryption uses the **same key** to encrypt and decrypt, whereas asymmetric encryption uses a **pair of keys** — a **public key** to encrypt and the matching **private key** to decrypt (1 — for the key difference). Symmetric is **faster** but suffers the **key-distribution problem** (the shared key must be transmitted securely) (1). Advantage of asymmetric (any one): the public key can be **shared openly**, which solves the key-distribution problem; it also enables **digital signatures** (sign with private key, verify with public key) (1). *(AO1)*

8. [3] Round robin gives each process a **fixed time slice (quantum)** of CPU time in turn (1). If a process does not finish within its slice it is **paused and moved to the back of the queue**, and the next process runs; this cycle repeats until all processes complete (1 — it is pre-emptive). Advantage (any one): it is **fair** — every process gets regular CPU time, so **no process starves**, and it gives good responsiveness for interactive tasks (1). *(AO1)*